

33

Diastereoselectivity

Connections

➡ Building on

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Arriving at

- How to make single diastereoisomers from geometrical isomers
- How to predict and explain reactions of chiral carbonyl compounds
- How chelation to metal ions can alter stereoselectivity
- How to predict and explain the reactions of chiral alkenes
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Looking back

You have had two chapters in a row about stereochemistry: this is the third, and it is time for us to bring together some ideas from earlier in the book. We aim firstly to help you grasp some important general concepts and secondly to introduce some principles in connection with stereoselective reactions in acyclic systems. But, first, some revision.

We introduced the stereochemistry of structures in Chapter 14. We told you about two types of stereoisomers.

● Enantiomers and diastereoisomers

- **Enantiomers**—stereoisomers that are mirror images of one another.
- **Diastereoisomers**—stereoisomers that are not mirror images of one another.

In this chapter we shall talk about how to make compounds as single *diastereoisomers*. Making single *enantiomers* is treated in Chapter 41. Chapter 32 was also about making single diastereoisomers, and we hope that, having read that chapter, you are used to thinking stereochemically. We shall meet two different ways of making single diastereoisomers.

● Reactions that make single diastereoisomers

- **Stereospecific reactions**—reactions where the mechanism means that the stereochemistry of the starting material determines the stereochemistry of the product and there is no choice involved.
- **Stereoselective reactions**—reactions where one stereoisomer of product is formed predominantly because the reaction has a choice of pathways, and one pathway is more favourable than the other.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

These terms were introduced in Chapter 17 in connection with elimination reactions, and many of the reactions we mention will be familiar from earlier chapters (particularly Chapters 15–19, 25, and 26). A common misapprehension is that stereospecific means merely very stereoselective. It doesn't—the two terms describe quite different properties of the stereochemistry of a reaction. For the purposes of making a single diastereoisomer, you can think of stereospecific reactions as ones which simply exchange different forms of stereochemical 'currency' (double bond geometry and three-dimensional relative stereochemistry, for example) while stereoselective reactions create additional new stereochemical value.

Making single diastereoisomers using stereospecific reactions of alkenes

The essence of the definition we have just stated is much easier to grasp with some familiar examples. Here are two.

- S_N2 reactions are stereospecific: they proceed with inversion so that the absolute stereochemistry of the starting material determines the absolute stereochemistry of the product.



➡ This is discussed in Chapter 15, p. 344.

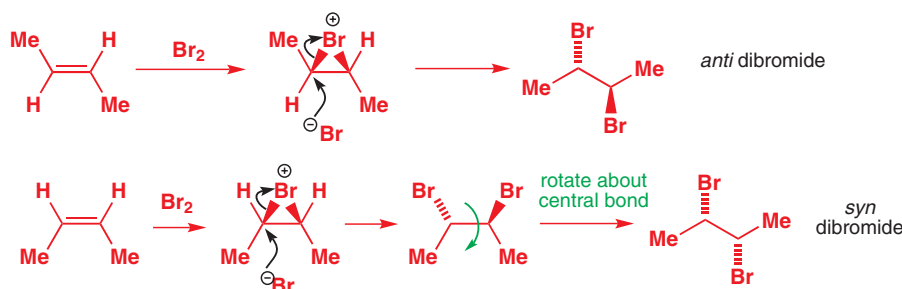
- E2 reactions are stereospecific: they proceed through an anti-periplanar transition state, with the relative stereochemistry of the starting material determining the geometry of the product.



➡ This is discussed in Chapter 17, p. 395.

Both of these examples are interesting because they show how, once we have some stereochemistry in a molecule, we can change the functional groups but keep the stereochemistry—this is the essence of a stereospecific reaction. In the second example, we change the bromide to a double bond, but we keep the stereochemistry (or 'stereochemical information') because the geometry of the double bond tells us which bromide we started with.

This is a good place to begin if we want to make single diastereoisomers because we can reverse this type of reaction: instead of making a single geometry of alkene from a single diastereoisomer, we make a single diastereoisomer from a single geometry of double bond. Here is an example of this—again, one you have already met (Chapter 19). Electrophilic addition of bromine to alkenes is stereospecific and leads to *anti* addition across a double bond. So if we want the *anti* dibromide we choose to start with the *trans* double bond; if we want the *syn* dibromide we start with the *cis* double bond. The geometry of the starting material determines the relative stereochemistry of the product.

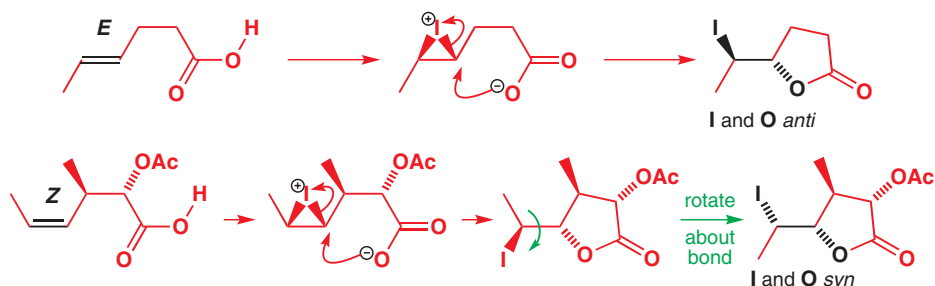


➡ Chapter 27 described the methods available for controlling the geometry of double bonds.

🔗 Interactive mechanism for stereospecific *anti* addition to alkenes

Iodolactonization has a similar mechanism; notice how in these two examples the geometry of the double bond in the starting material defines the relative stereochemistry highlighted in black in the product.

■ There are two more stereogenic centres in the second example here and, although they do not affect the relative stereochemistry shown in black, they do affect how those two new stereogenic centres relate to the two that are already present in the starting material. We discuss how later in the chapter.



For a stereospecific alkene transformation, choose the right geometry of the starting material to get the right diastereoisomer of the product. Don't try to follow any 'rules' over this—just work through the mechanism.

Now for some examples with epoxides. Epoxides are very important because they can be formed stereospecifically from alkenes: *cis* alkenes give *cis* (or *syn*) epoxides and *trans* alkenes give *trans* (or *anti*) epoxides.

➔ Formation and reaction of epoxides are described in Chapter 19, p. 429 and Chapter 15, p. 351.



Epoxides also *react* stereospecifically because the ring-opening reaction is an S_N2 reaction. A single diastereoisomer of epoxide gives a single diastereoisomer of product.



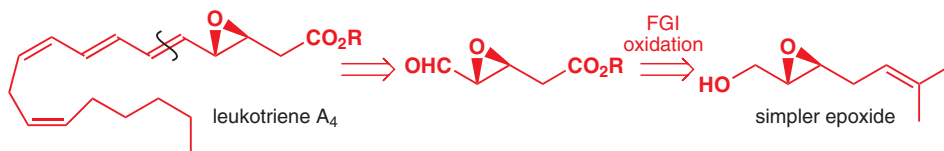
Leukotrienes

Leukotrienes are metabolites of arachidonic acid related to prostaglandins and thromboxanes. They are made in nature (and often in the laboratory) by oxidation of alkenes. The letter (e.g. A) gives the general structure and the subscript number the number of alkenes. They are unstable and control localized physiological phenomena such as blood clotting and inflammation.

Leukotrienes are important molecules that regulate cell and tissue biology. Leukotriene C_4 (LTC_4) is a single diastereoisomer with an *anti* 1,2 S,O functional group relationship. In nature, this single diastereoisomer is made by an epoxide opening: since the opening is S_N2 the epoxide must start off *anti* and, indeed, the epoxide precursor is another leukotriene, LTA_4 .

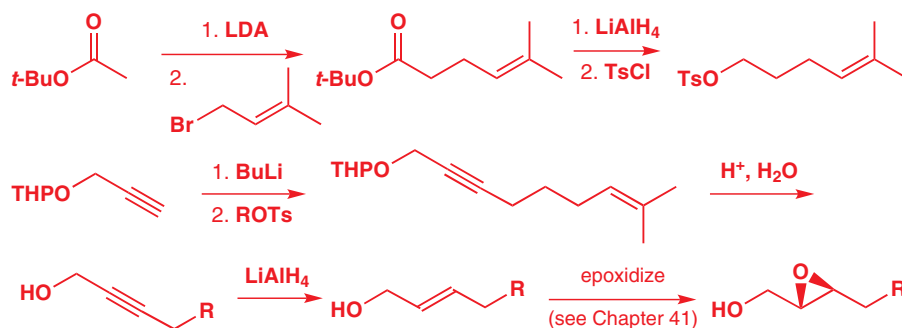


When Corey was making these compounds in the early 1980s he needed to be sure that the relative stereochemistry of LTC_4 would be correctly controlled, and to do this he had to make a *trans* epoxide. Disconnecting LTA_4 led back to a simpler epoxide.



The *trans* allylic alcohol needed to make this compound was made using one of the methods we introduced in Chapter 27: reduction of an alkynyl alcohol with $LiAlH_4$. Here is the full synthesis: alkylation of an ester enolate with prenyl bromide gives a new ester, which itself is turned into an alkylating agent by reduction and tosylation. The alkyne is introduced as its lithium derivative with the alcohol protected as a THP acetal. Hydrolysis of the acetal with

aqueous acid gives the hydroxy-alkyne needed for reduction to the *E* double bond, which is then epoxidized.

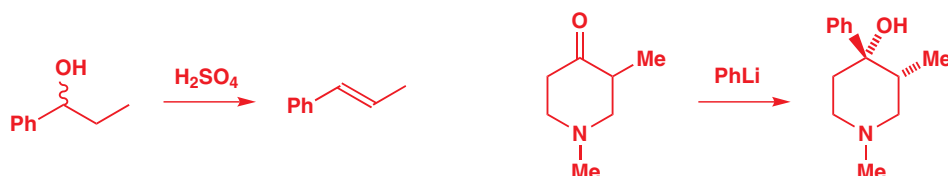


➡ The epoxide was, in fact, made as a single enantiomer using the Sharpless epoxidation, which we will describe in Chapter 41.

Stereoselective reactions

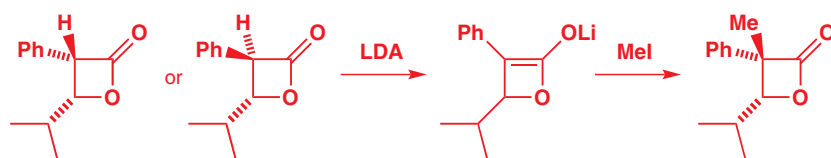
For most of the rest of the chapter we shall discuss stereoselective reactions. You have already met several examples and we start with a summary of the most important methods.

- E1 reactions are stereoselective: they form predominantly the more stable alkene.
- Nucleophilic attack on six-membered ring ketones is stereoselective: small nucleophiles attack axially and large ones equatorially.



➡ Chapter 17, p. 391 and Chapter 32, p. 829.

- Alkylation of cyclic enolates is stereoselective, with reaction taking place on the less hindered face (four- or five-membered rings) or via axial attack (six-membered rings).



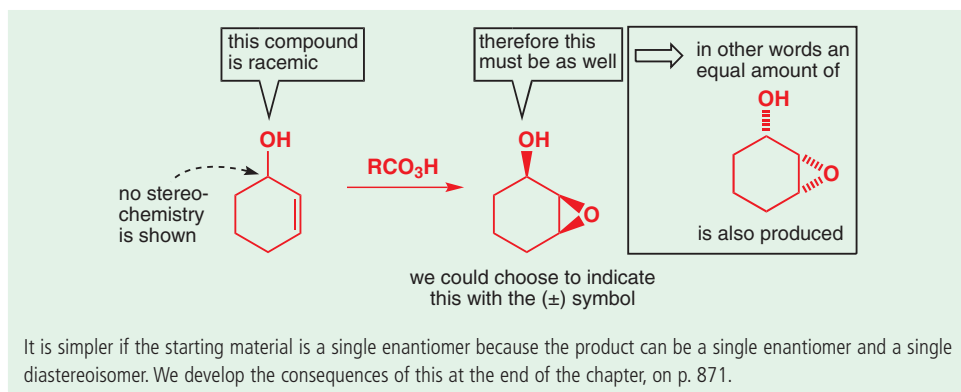
➡ Chapter 32, pp. 833 and 850.

- Epoxidation of cyclic alkenes is stereoselective, with reaction taking place on the less hindered face, or directed by hydrogen bonding to a hydroxyl group.



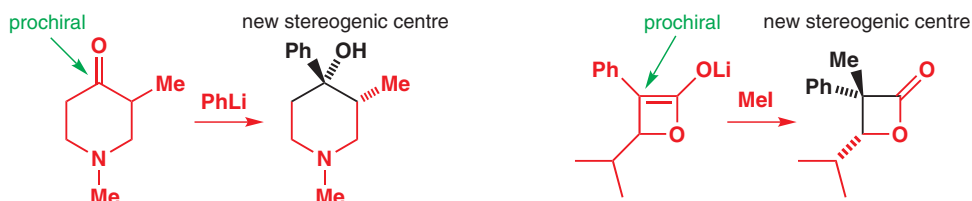
■ A comment on structural drawings of single diastereoisomers

In the two reactions just above, a racemic starting material gives a racemic product as a single diastereoisomer. It is easy to draw a racemic compound with just one stereogenic centre—we just avoid showing stereochemistry. But in the products we *have* to show relative stereochemistry because we need to indicate which diastereoisomer is formed. There is no way of doing this except by arbitrarily choosing one enantiomer and drawing that. If there is a danger of confusion, we might sometimes write '(±)' under the structure.

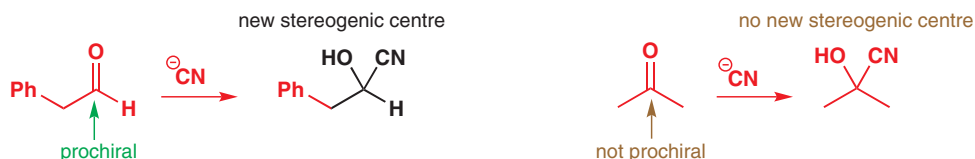


Prochirality

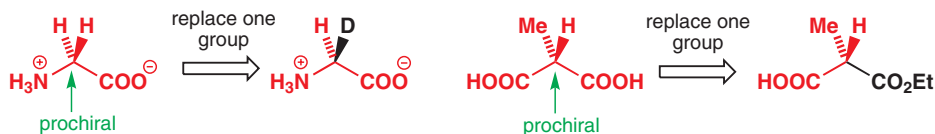
Take another look at the reactions in the chapter so far—in particular those that give single diastereoisomers (rather than single enantiomers or geometrical isomers), in other words, those that are diastereoselective. They all involve the creation of a new, tetrahedral stereogenic centre at a carbon that was planar and trigonal. This leads us to our first new definition. Trigonal carbons that aren't stereogenic (or chiral) centres but can be made into them are called *prochiral*.



At the very start of Chapter 15 we introduced stereochemistry by thinking about the reactions of two sorts of carbonyl compounds. They are shown again here: the first has a prochiral carbonyl group. The second, on the other hand, is not prochiral because no stereogenic centre is created when the compound reacts.



A tetrahedral carbon atom can be prochiral too—if it carries two identical groups (and so is not a chiral centre) but replacement of one of them leads to a new chiral centre, then the carbon is prochiral.

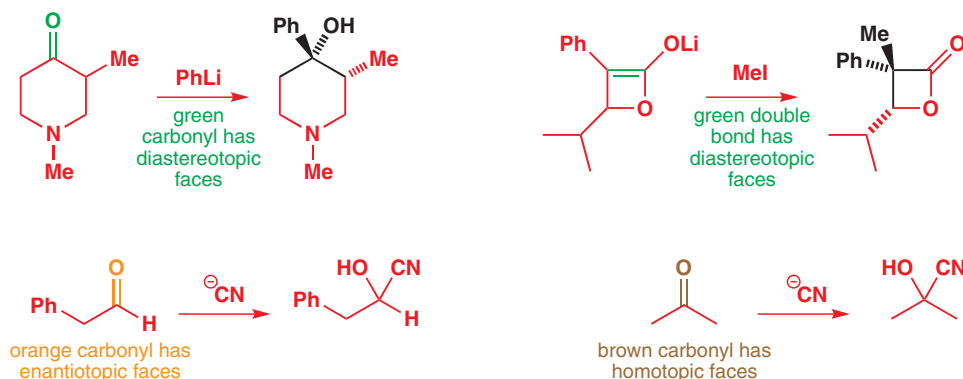


➡ Enantiotopic and diastereotopic protons and groups are discussed in Chapter 31, p. 820.

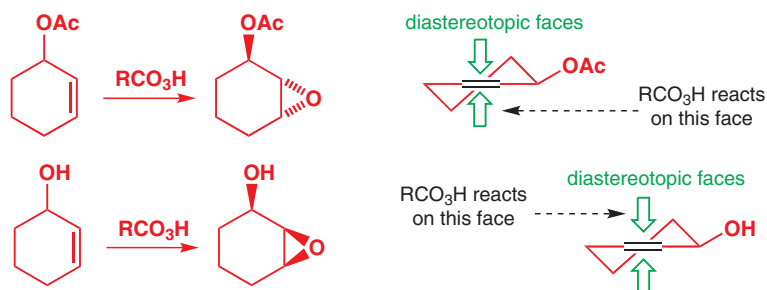
Glycine is the only common α amino acid without a chiral centre, but replacing one of the two protons on the central carbon with, say, deuterium creates one: the CH_2 carbon is prochiral. Similarly, converting a malonate derivative into its monoester makes a chiral centre where there was none: the central C is prochiral. Now, does this ring any bells? It should remind you very much of the definitions in Chapter 31 of *enantiotopic* and *diastereotopic* in connection with NMR spectra. Replacing one of two enantiotopic groups with another group leads to one of two enantiomers; replacing one of two diastereotopic groups with another group leads to

one of two diastereoisomers. Diastereotopic groups are chemically different; enantiotopic groups are chemically identical.

Exactly the same things are true for the faces of a prochiral carbonyl group or double bond. If reaction on one of two faces of the prochiral group generates one of two enantiomers, the faces are enantiotopic; if the reaction generates one of two diastereoisomers, the faces are diastereotopic. We will now apply this thinking to the first few reactions in this chapter: they are shown again below. The two examples in the top row have prochiral C=C or C=O bonds with diastereotopic faces: choosing which face of the double bond or carbonyl group to react on amounts to choosing which diastereoisomer to form. In the third example, the faces of the prochiral carbonyl group are enantiotopic: choosing which face to attack amounts to choosing which enantiomer to form. In the fourth example, the two faces of C=O are **homotopic**: an identical product is formed whichever face is attacked.



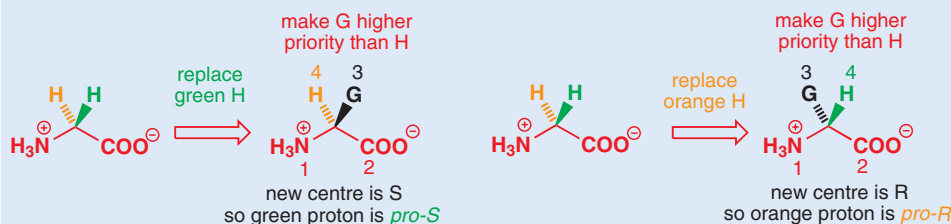
Knowing this throws some new light on the last chapter. Almost without exception, every stereoselective reaction there involved a double bond (usually C=C, sometimes C=O) with diastereotopic faces. The diastereotopic faces were distinguished by steric hindrance, or by a nearby hydrogen-bonding group, and so were able to react differently with an incoming reagent.



Using an *R/S*-type system to name prochiral faces and groups

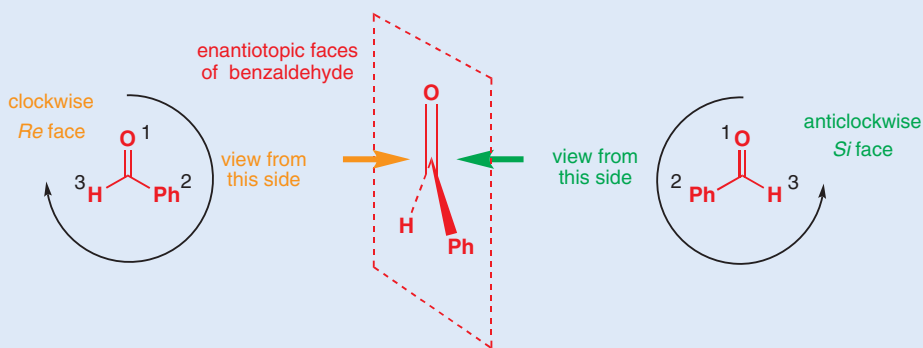
Just as stereogenic centres can be described as *R* or *S*, it is possible to assign labels to the enantiotopic groups at prochiral tetrahedral carbon atoms or the enantiotopic faces of prochiral trigonal carbon atoms. The basis of the system is the usual *R, S* system for stereogenic centres, but *pro-R* and *pro-S* are used for groups and *Re* and *Si* for faces.

Pro-R and *pro-S* can be assigned to a pair of enantiotopic groups simply by using the usual rules to assign *R* or *S* to the centre created *if* the group in question is artificially elevated to higher priority than its enantiotopic twin. We'll use G to replace H as we did in Chapter 31: just assume that G has priority immediately higher than H. The method is illustrated for glycine.



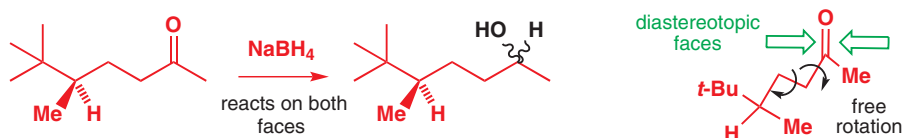
Faces of a prochiral trigonal carbon atom are assigned *Re* and *Si* by viewing the carbon from that side and counting down the groups in priority 1–3. Counting round to the right (clockwise) means the face is *Re*; counting round to the left (anticlockwise) means it's *Si*. Remember our advice from Chapter 14: think of turning a steering wheel in the direction of the numbers: does the car go to the right or the left?

Like *R* and *S*, these stereochemical terms are merely labels: they are of no consequence chemically.



■ In Chapter 32 we showed that homotopic and enantiotopic protons are identical by NMR. Similarly homotopic faces or groups are always chemically identical. Enantiotopic faces are also chemically identical provided that all the reagents in the reaction in question are achiral or racemic. In Chapter 41 we will consider what happens to enantiotopic faces when *enantiomerically pure* reagents are used.

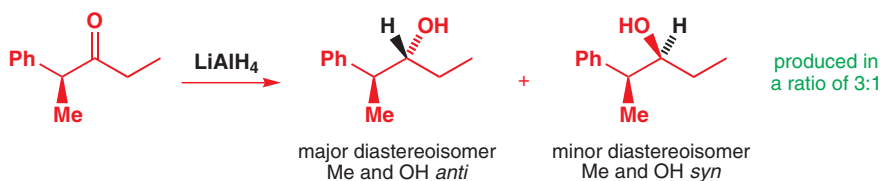
Just like diastereotopic signals in an NMR spectrum, diastereotopic faces are always different in principle, but sometimes not so in practice. The very first reaction of Chapter 32 is a case in point: this C=O group has two diastereotopic faces, which, due to free rotation about single bonds, average out to about the same reactivity, so we cannot expect any reasonable level of diastereoselectivity.



We put Chapter 32 first because in rings conformation is well defined, and this 'averaging' effect is held at bay. We are about to let it out again, but we will show you how it can be tamed to surprisingly good effect.

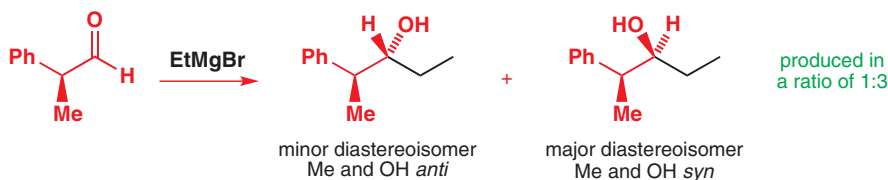
Additions to carbonyl groups can be diastereoselective even without rings

What happens if we bring the stereogenic centre closer to the carbonyl group than it was in the last example? You might expect it to have a greater influence over the carbonyl group's reactions. And it does. Here is an example.



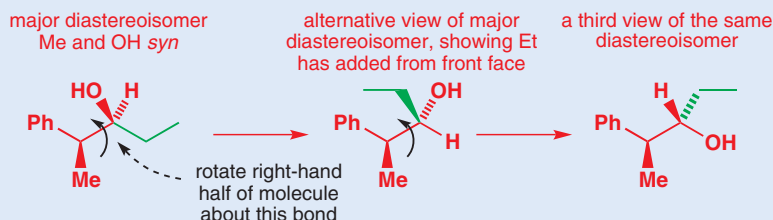
■ We have termed the major diastereoisomer *anti* because the two substituents (Me and OH) are on opposite sides of the chain as drawn. There is no formal definition of *anti* and *syn*: they should be used only in conjunction with a structural drawing.

There is three times as much of one of the two diastereoisomeric products as there is of the other, and the major (*anti*) diastereoisomer is the one in which the nucleophile has added to the front face of the carbonyl group as drawn here. We can make these same two diastereoisomers by addition of an organometallic to an aldehyde. For example, this Grignard reagent gives three times as much of the *syn* diastereoisomer as the *anti* diastereoisomer. The major product has changed, but the product still arises from attack on the front face of the carbonyl as shown.



Drawing diastereoisomers of acyclic molecules

The three structures below all show the same diastereoisomer (the major product from the last reaction), but in three different conformations (we are just rotating about a bond to get from one to another).



Which is the best? A good guideline, which we suggested in Chapter 14, is to place the longest carbon chain zig-zagging across the page in the plane of the paper, and allow all the smaller substituents to extend above or below that chain. The first structure here is drawn like that. But this is only a guideline, and the second structure here is a bit more informative regarding the reaction because, when it is drawn like this, you can clearly see from which direction the ethyl group has attacked the carbonyl. Our advice would be that you first of all draw the product of any reaction in more or less the same conformation as the starting material to ensure you make no mistakes, and then rotate about a single bond to place the longest chain in the plane of the paper.

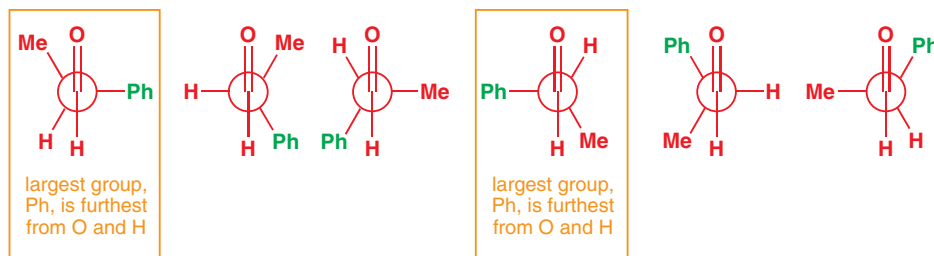
If you still have problems manipulating structures mentally—for example, if you find it hard to work out whether the substituents that aren't in the plane should be in front of or behind the page—build some models.

These two reactions are not nearly as diastereoselective as most of the reactions of cyclic compounds you met in the last chapter. But we do now need to explain why they are diastereoselective at all, given the free rotation possible in an acyclic molecule. The key, as much with acyclic as with cyclic molecules, is **conformation**.

The conformation of a chiral aldehyde

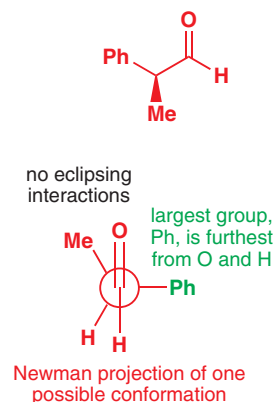
What will be the conformation of the aldehyde in the margin? Using the principles we outlined in Chapter 16, we can expect it to be staggered, with no eclipsing interactions, and also with large substituents as far apart from one another as possible. A Newman projection of one of the possible conformers might look like the one shown in the margin. There are no eclipsing interactions, and the large phenyl group is held satisfactorily far away from the O and the H atoms of the aldehyde.

By rotating about the central bond of the aldehyde (the one represented by a circle in the Newman projection) we can suggest a series of possible conformations. Provided we move in 60° steps, none of them will have any eclipsing interactions. The full set of six conformers is shown here. Look at them for a moment and notice how they differ.



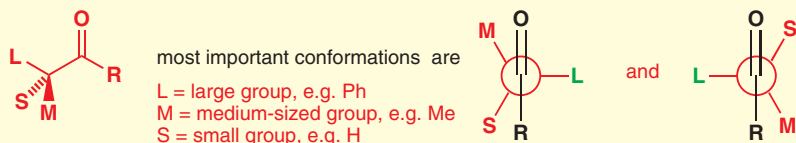
Only two of them, boxed in orange, place the large Ph group perpendicular to the carbonyl group. These yellow-boxed conformations are therefore the lowest-energy conformers and, for the purpose of the discussion that follows, they are the only ones whose reactions we need to consider.

➡ We shall draw heavily on the first part of Chapter 16: if you haven't read it recently, now might be a good time to refresh your memory.



• Lowest-energy conformations of a carbonyl compound

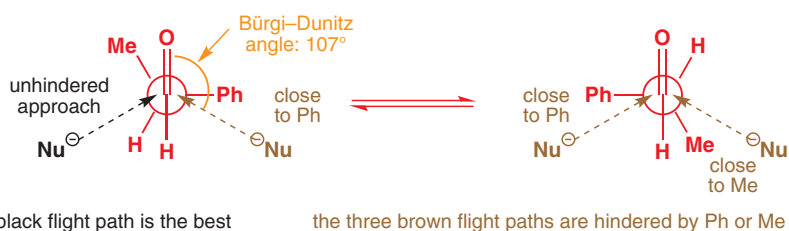
The most important conformations of a carbonyl compound with a stereogenic centre adjacent to the carbonyl group are those that place the largest group perpendicular to the carbonyl group.



The major product arises from the most reactive conformer

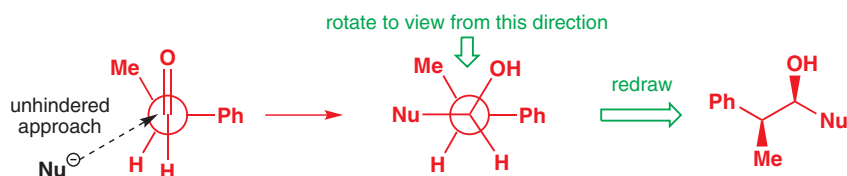
Now that we have decided which are the important conformations, how do we know which gives the product? We need to decide which is the *most reactive*. All we need to do is to remember that any nucleophile attacking the carbonyl group will do so from the Bürgi–Dunitz angle—about 107° from the C=O bond. The attack can be from either side of C=O, and the following diagrams show the possible trajectories superimposed on the two conformations we have selected, which are in equilibrium with one another.

➡ We introduced the idea that attack on a C=O group followed this trajectory in Chapter 6 (p. 127).



This is an example of the **Curtin–Hammett principle**, which says that it is the relative energies of the transition states that control selectivity, not the relative energies of the starting materials. It's really more of a reminder not to make a mistake than a principle.

Not all four possible 'flight paths' for the nucleophile are equally favourable. For the three shown in brown, the nucleophile passes within 30° or so of another substituent. But, for the one shown in black, there is no substituent nearby except H to hinder attack: the conformation on the left is the most reactive one, and it reacts to give the diastereoisomer shown below.



Cram's rule

You may hear 'Cram's rule' used to explain the outcome of reactions involving attack on chiral carbonyl compounds. Cram was the first to realize that these reactions could be predicted, but we now know why these compounds react in a predictable way. We will not describe Cram's rule because, although it often does predict the right product, in this case it does so for the wrong reason. Explanations and clear logical thinking are more important than rules, and you must be able to account for and predict the reactions of chiral aldehydes and ketones using the Felkin–Anh model.

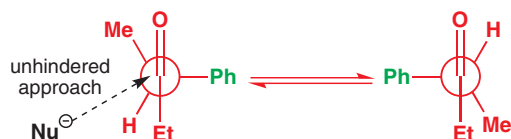
• In order to avoid making mistakes, we suggest you:

- first draw the product in a conformation similar to that of the starting material
- then redraw to put the longest chain in the plane of the paper.

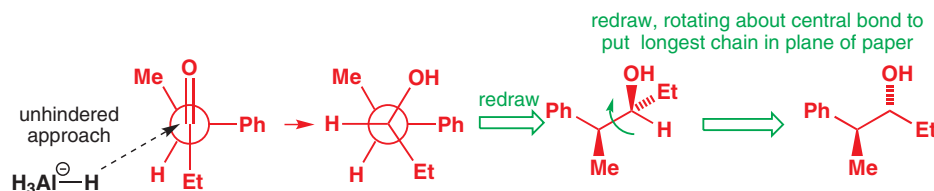
Here, this just means drawing the view from the top of the Newman projection—there is no need to rotate any bonds in this case.

With Nu=Et we have the right product and, more importantly, we can be pretty sure it is for the right reason: this model of the way a nucleophile attacks a carbonyl compound, called the **Felkin–Anh model**, is supported by theoretical calculations and numerous experimental results. Notice that we don't have to decide which is the lower energy of the two conformations: this is not necessary because the attack in black will occur even if the conformer on the left is the minor one in the mixture.

The same reasoning accounts for the diastereoselectivity of the reduction on p. 858: first we need to draw the two important conformers of the ketone; the ones that have the large group (Ph) perpendicular to the C=O group.



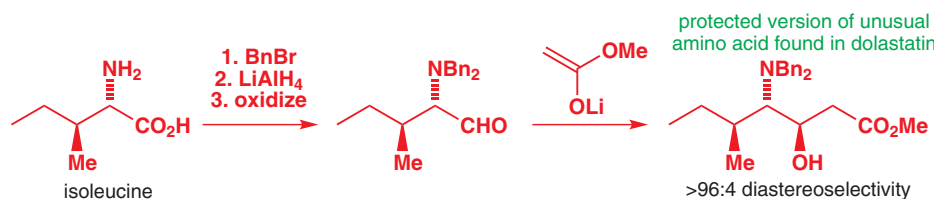
Now choose the angle of attack that is the least hindered and draw a Newman projection of the product. Finally, redraw the Newman projection as a normal structure, preferably with the longest chain in the plane of the paper.



Interactive Felkin–Anh model for ketone reduction

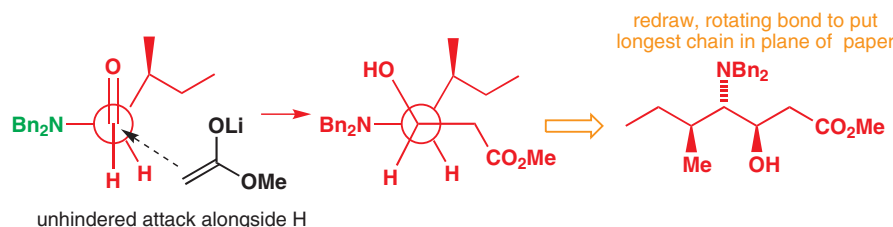
The effect of electronegative atoms

One of the most powerful anticancer agents known is dolastatin, isolated from the sea-hare *Dolabella*. Dolastatin contains an unusual amino acid, with three stereogenic centres, and chemists in Germany managed to exploit Felkin–Anh control very effectively to make it from the much more widespread amino acid isoleucine. This is the sequence of reactions.



■ When you see a selectivity given as 'greater than' something it means that the other diastereoisomer was undetectable and here 96:4 was the limit of detection by the method used—possibly NMR.

The key step is the aldol reaction of the enolate of methyl acetate with the protected amino aldehyde. To rationalize the stereoselectivity, we first need to draw the two most important conformations of this aldehyde with the large group perpendicular to C=O. The trouble is, which do we choose as 'large': the NBn₂ group or the branched alkyl group? Since we know which diastereoisomer is produced we can work backwards to find that it must be the NBn₂ group that sits perpendicular to C=O in the reactive transition state, and not alkyl. We can draw the best conformation without worrying about alternatives.



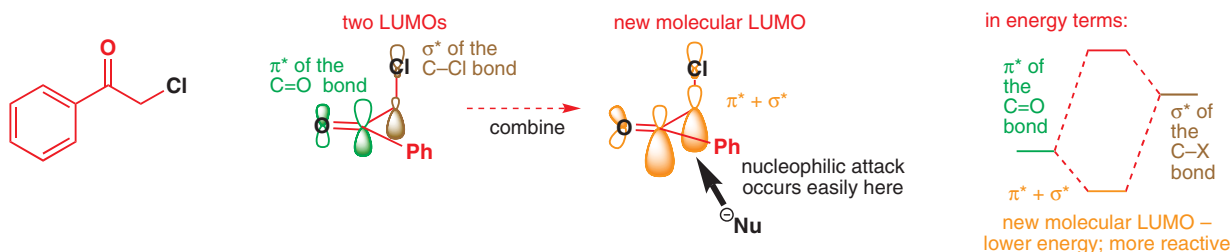
■ Try for yourself putting alkyl perpendicular to C=O: you will get the wrong diastereoisomer.

Now look at the diastereoselectivity of the reaction: it is much greater than the 3:1 we saw before—more like 20:1. This really does suggest that there is a further factor at work here, and that further factor is the electronegative N atom.

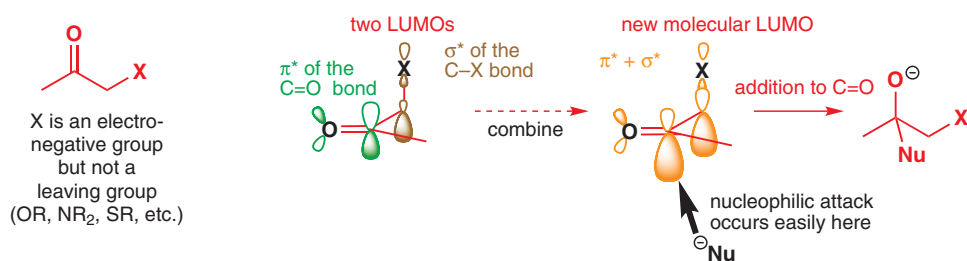
Carbonyl groups increase the reactivity of adjacent leaving groups towards nucleophilic substitution by several orders of magnitude. This was an effect that we noted in Chapter 15, where we showed that the ketone below reacts by the S_N2 mechanism 5000 times as fast as methyl chloride itself. We explained this effect by saying that the π* of the C=O and the σ* of C–Cl overlap to form a new, lower-energy (and therefore more reactive) LUMO. What we did not

➡ This is discussed on p. 341 of Chapter 15.

note then, because it was not relevant, is that this overlap can only occur when the C–Cl bond is perpendicular to the C=O bond, because only then are the π^* and σ^* orbitals aligned correctly.



The same thing happens even with electronegative atoms X that are not leaving groups in the $\text{S}_{\text{N}}2$ reaction (for example, X=OR, NR_2 , SR, etc.). The π^* and σ^* orbitals add together to form a new, lower-energy molecular orbital, more susceptible to nucleophilic attack. But, if X is not a leaving group, attack on this orbital will result not in nucleophilic substitution but in addition to the carbonyl group. Again, this effect will operate only when the C–X and C=O bonds are perpendicular so that the orbitals align correctly.



What does this mean for stereoselectivity? Conformations of the chiral carbonyl compound that place an electronegative atom perpendicular to the C=O bond will be more reactive—size doesn't matter. So, in the dolastatin amino acid example, the conformations with NBn_2 perpendicular to C=O are the only conformations we need to consider.

● Using the Felkin–Anh model

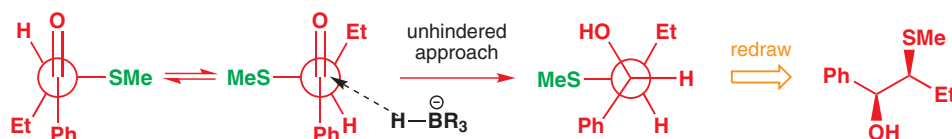
To predict or explain the stereoselectivity of reactions of a carbonyl group with an adjacent stereogenic centre, use the Felkin–Anh model. If you look at the next example, just below this box, you can follow exactly the series of steps we suggest you take:

- Draw Newman projections of the conformations of the starting material that place a large group or an electronegative group perpendicular to C=O.
- Allow the nucleophile to attack along the least hindered trajectory, taking into account the Bürgi–Dunitz angle.
- Draw a Newman projection of the product that arises from attack in this way.
- Carefully flatten the Newman projection on to the page to produce a normal structure, preferably with the longest chain of C atoms in the plane of the page. Check that you have done this last step correctly: it is very easy to make mistakes here. Use a model if necessary, or do the 'flattening out' in two stages—first view the Newman projection from above or below and draw that; then rotate some of the molecule about a bond if necessary to get the long chain into the plane of the page.

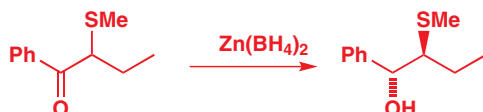
Chelation can reverse stereoselectivity



You should now be in a position to explain the outcome of this reaction without much difficulty. Sulfur is the electronegative atom, so the conformations we need to consider are the two following. Unhindered attack on the second gives the diastereoisomer shown.

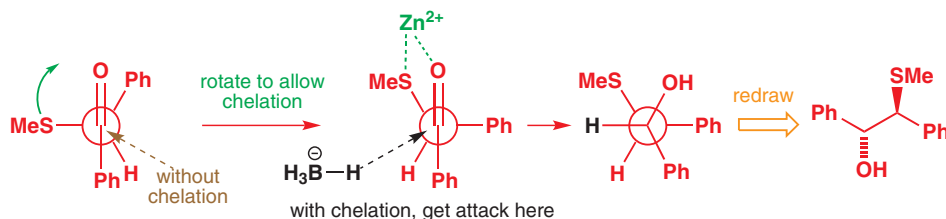


But, from what we have told you so far, the next reaction would present a problem: changing the metal from lithium to zinc has reversed the stereoselectivity. Using the simple Felkin–Anh model no longer works: it gives the wrong answer.



The reason is that zinc can chelate sulfur and the carbonyl group. Chelation is the coordination of two heteroatoms carrying lone pairs to the same metal atom, and here it changes the conformation of the starting material. No longer does the most reactive or most populated conformation place the electronegative S atom perpendicular to C=O; instead it prefers S to lie as close to the carbonyl oxygen as possible so that Zn can bridge between S and O, like this:

A chelate (from the Greek for ‘claw’) is a coordination compound with a metal atom bonded to an organic molecule at two or more atoms.



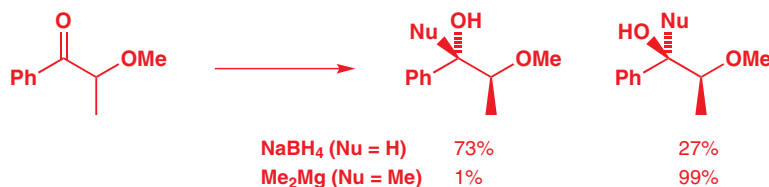
Interactive model for chelation-controlled ketone reduction

When chelation is possible, this is the conformation to consider—the one with the carbonyl O and the other chelating atom almost eclipsing one another. It is the most populated because it is stabilized by the chelation, and it is also the most reactive, because the Lewis-acidic metal atom increases the reactivity of the carbonyl group. Attack is still along the less hindered pathway, but this now leads to the other face of the carbonyl group, and the stereochemical outcome is reversed.

Two things are needed for chelation to occur:

- a heteroatom with lone pairs available for coordination to a metal
- a metal ion that prefers to coordinate to more than one heteroatom at once—these are mainly more highly charged ions, as shown in the table.

Here is another example of a reversal in selectivity that can be explained using a non-chelated Felkin–Anh model with Na^+ and a chelated model with Mg^{2+} .



Metals commonly involved in chelation	Metals not usually involved in chelation
Li^+ often	Na^+
Mg^{2+}	K^+
Zn^{2+}	
Cu^{2+}	
Ti^{4+}	
Ce^{3+}	
Mn^{2+}	

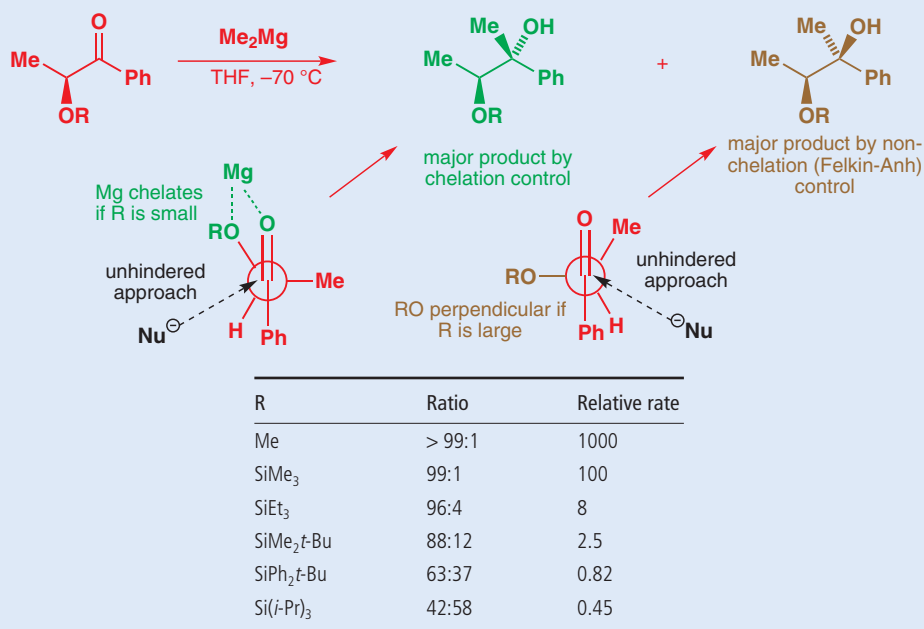
Not only does chelation control reverse the stereoselectivity, it gives a much higher degree of stereoselectivity. Stereoselectivities in chelation-controlled additions to C=O groups are typically >95:5. But this fits in nicely with the ideas we presented at the end of the last chapter: stereoselectivity is likely to be high if a cyclic transition state is involved. Chelation involves

Interactive model for chelation-controlled organo-Mg addition

just such a transition state, so it should be no surprise that it lets us achieve much higher levels of control than the acyclic Felkin–Anh model does.

Chelation, rate, and stereoselectivity

The correlation of rate of addition with diastereoselectivity was demonstrated in a series of experiments that involved reacting Me_2Mg with protected α -hydroxy-ketones. As the protecting group was changed from a methyl ether to a trimethylsilyl ether and then through a series of increasingly bulky silyl ethers, both the rate of the reaction and the diastereoselectivity decreased. With small protecting groups the reaction takes place through the chelated transition state—the selectivity shows this—and the rate is faster because of the activating effect of the Lewis-acidic magnesium ion. But with larger protecting groups chelation of Mg^{2+} between the two oxygen atoms is frustrated: the rate drops off and the selectivity becomes more what would be expected from the Felkin–Anh model.

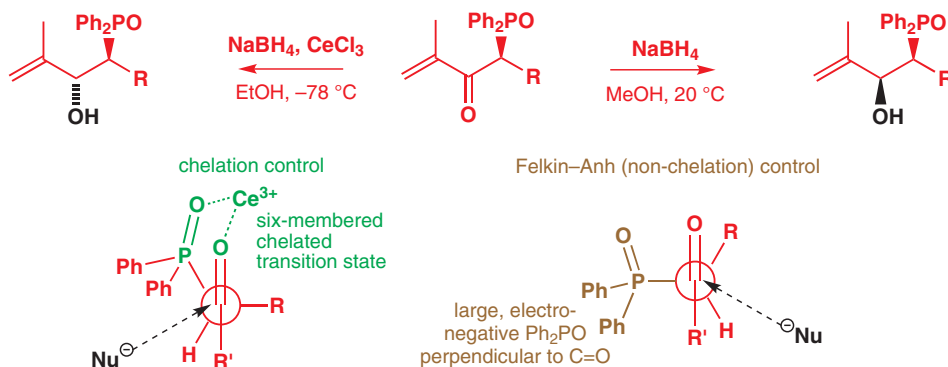


• Chelation:

- may change the direction of diastereoselectivity
- leads to high levels of diastereoselectivity
- increases the rate of the addition reaction.

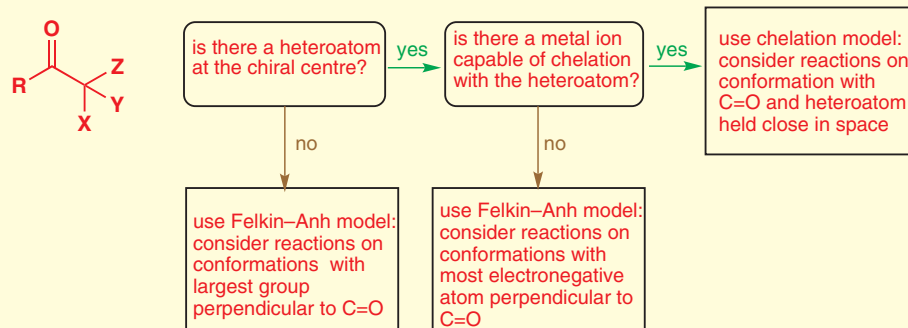
Chelation is possible through six- as well as five-membered rings and the reduction of the ketone below is a nice example of the reversal of diastereoselectivity observed when chelating Ce^{3+} ions are added to a normal sodium borohydride reduction. The products were important for making single geometrical isomers of alkenes in a modification of the Wittig reaction. Notice too that the rate changes: with Ce^{3+} the reaction can be done at -78°C .

➡ This is Luche reduction, mentioned in Chapter 23, p. 536.



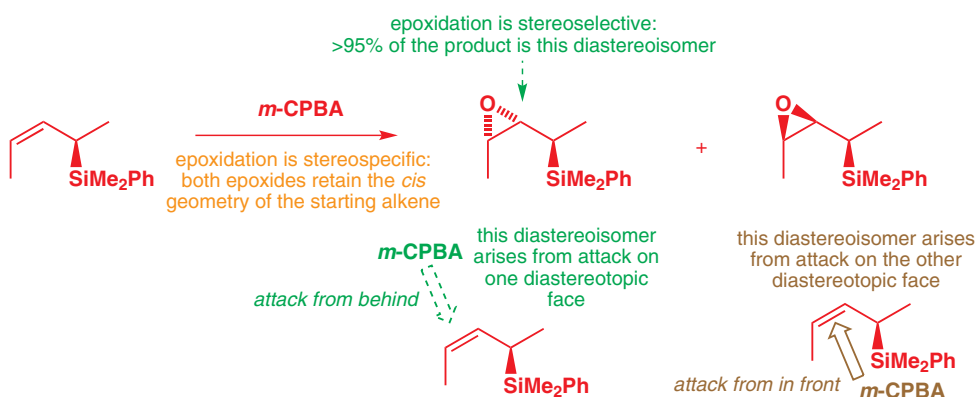
● Attack on α chiral carbonyl compounds: summary

The flow chart summarizes what you should consider when you need to predict or explain the stereochemical outcome of nucleophilic attack on a chiral carbonyl compound.



Stereoselective reactions of acyclic alkenes

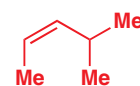
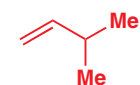
Earlier in the chapter we discussed how to make single diastereoisomers by stereospecific additions to double bonds of fixed geometry. But if the alkene also contains a chiral centre there will be a stereoselective aspect to its reactions too: its faces will be diastereotopic, and there will be two possible outcomes even if the reaction is fully stereospecific. Here is an example where the reaction is an epoxidation.



The Houk model

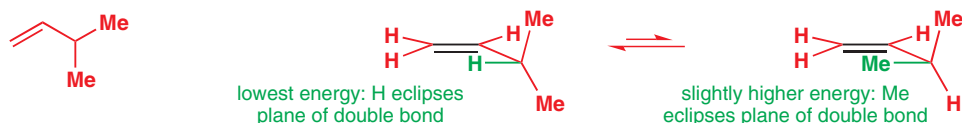
In order to explain reactions of chiral alkenes like this, we need to assess which conformations are important and consider how they will react, just as we have done for chiral carbonyl compounds. Much of the work on alkene conformations was done by K. N. Houk using theoretical computer models, and we will summarize the most important conclusions of these studies. The theoretical studies looked at two model alkenes, shown in the margin.

The calculations found that the low-energy conformations in each case were those in which a substituent eclipses the double bond. For the simple model alkene **1**, the lowest-energy conformation is the one that has the proton in the plane of the alkene. Another low-energy conformation—only 3.1 kJ mol⁻¹ higher—has one of the methyl groups eclipsing the double bond, so that when we start looking at reactions of this type of alkene, we shall have to consider both conformations.

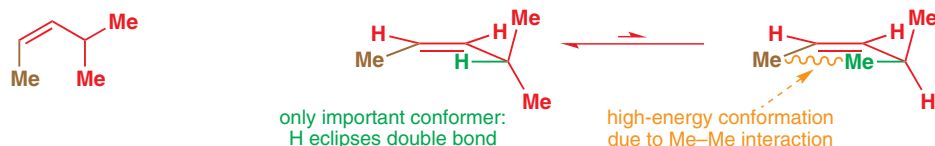


K. N. Houk works at the University of California in Los Angeles. He has provided explanations for a number of stereochemical results by using powerful computational methods.

model alkene 1 has two low-energy conformations



model alkene 2 has only one low-energy conformation



■ This effect—the control of conformation by a *cis* substituent—is known as **allylic strain** or $A^{1,3}$ strain because the groups involved are on carbons 1 and 3 of an allylic system.

For the model alkene 2, with a *cis* substituent, the conformation is more predictable and the only low-energy conformer is the one with the hydrogen eclipsing the double bond. There is no room for a methyl group to eclipse the double bond because if it did it would get too close to the *cis* substituent at the other end of the double bond.

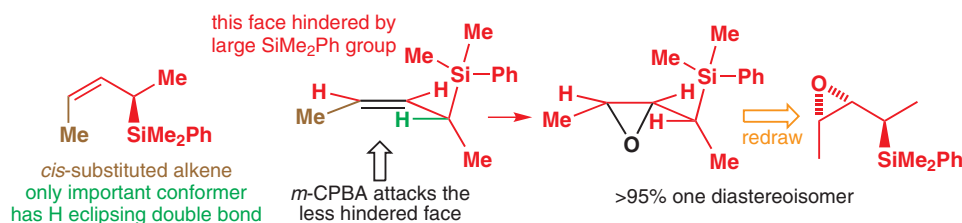
The message from the calculations is this:

- The lowest-energy conformation of a chiral alkene will have H eclipsing the double bond.
- If there is a *cis* substituent on the alkene, this will be the only important conformation; if there is no *cis* substituent, other conformations may be important too.

Now we can apply the theoretical model to some real examples.

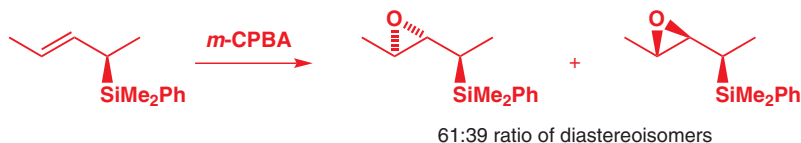
Stereoselective epoxidation

We started this section with a diastereoselective epoxidation of an alkene. The alkene was this one, and it has a substituent *cis* to the stereogenic centre. We can therefore expect it to have one important conformation, with H eclipsing the double bond. When a reagent—*m*-CPBA here—attacks this conformation, it will approach the less hindered face, and the outcome is shown.

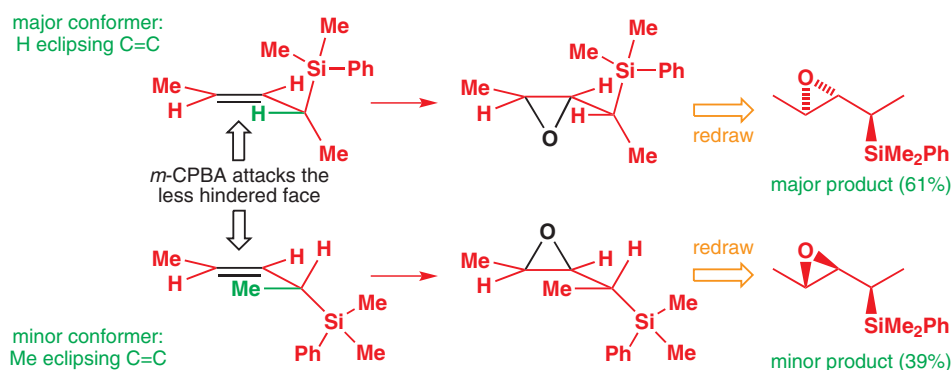


Interactive model for epoxidation controlled by allylic strain

Without the *cis* substituent, selectivity is much lower.



m-CPBA still attacks the less hindered face of the alkene, but with no *cis* substituent there are two low-energy conformations: one with H eclipsing the double bond, and one with Me eclipsing. Each gives a different stereochemical result, explaining the low stereoselectivity of the reaction.

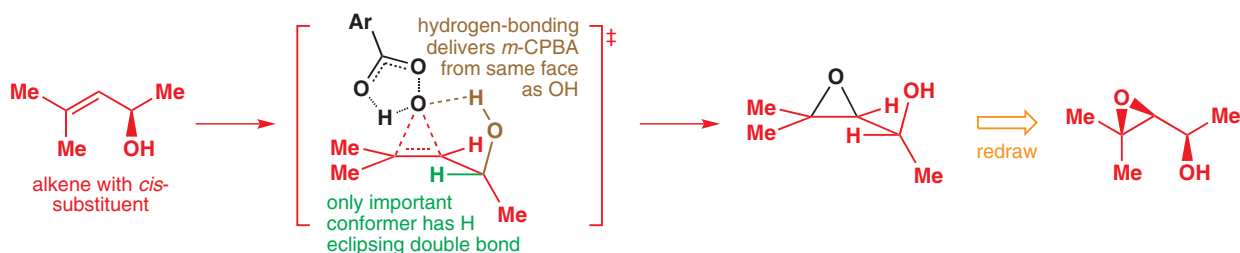


Again, draw the product in the same conformation as the starting material then flatten into the plane of the page.

You saw at the end of the last chapter that the reactions of *m*-CPBA can be directed by hydroxyl groups, and the same thing happens in the reactions of acyclic alkenes. This allylic alcohol epoxidizes to give a 95:5 ratio of diastereoisomers.



Drawing the reactive conformation explains the result. The thing that counts is the *cis* methyl group: the fact that there is a *trans* one too is irrelevant as it is just too far away from the stereogenic centre to have an effect on the conformation. The reaction uses a racemic mixture, but to explain the diastereoselectivity we just have to pick one enantiomer and show what happens to that.

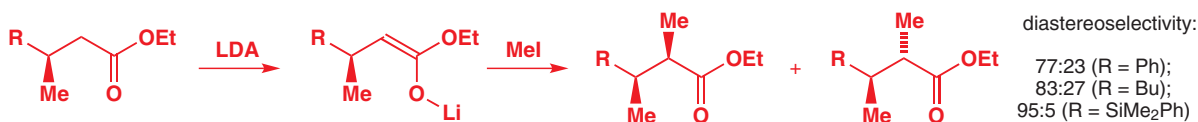


● To explain the stereoselectivity of reactions of chiral alkenes:

- Draw the conformation with H eclipsing the double bond.
- Allow the reagent to attack the less hindered of the two faces or, if coordination is possible, to be delivered to the face *syn* to the coordinating group.
- Draw the product in the same conformation as the starting material.
- Redraw the product as a normal structure with the longest chain in the plane of the paper.

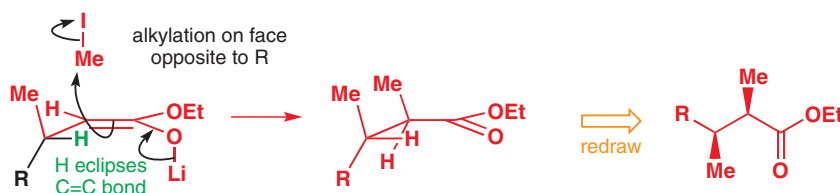
Stereoselective enolate alkylation

Chiral enolates can be made from compounds with a stereogenic centre β to a carbonyl group. Once the carbonyl is deprotonated to form the enolate, the stereogenic centre is next to the double bond and in a position to control the stereoselectivity of its reactions. The scheme below shows stereoselectivity in the reactions of some chiral enolates with methyl iodide.



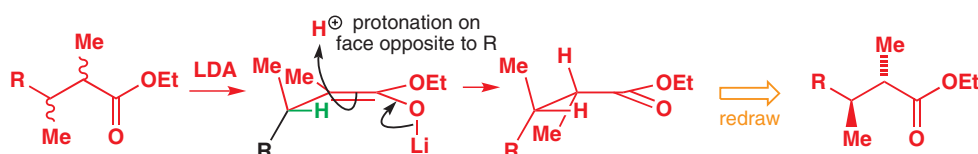
The enolate is a *cis*-substituted alkene because either O or OEt must be *cis* to the stereogenic centre, so that to explain the stereoselectivity we need consider only the conformation with H eclipsing the double bond. Notice how the diastereoselectivity increases as the group R gets bigger because there is then more contrast between the size of Me and R. In each case, the electrophile adds to the less hindered face, opposite R.

Interactive model for enolate alkylation controlled by allylic strain



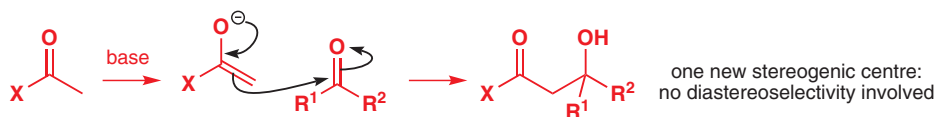
The other diastereoisomer can be made just by having the methyl group in place first and then protonating the enolate. The selectivities are lower (because a proton is small), but this does illustrate the way in which reversing the order of introduction of two groups can reverse the stereochemical outcome of the reaction.

■ The relative stereochemistry of the starting material is lost in the enolization step so either diastereoisomer or a mixture can be used.

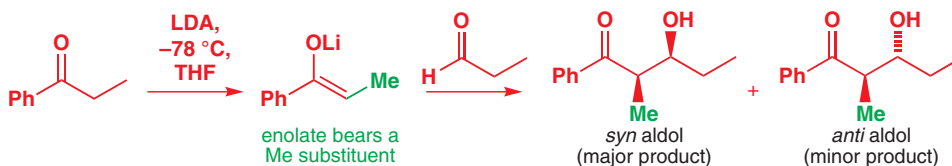


Aldol reactions can be stereoselective

In Chapter 26 you met the **aldol reaction**: reaction of an enolate with an aldehyde or a ketone. Many of the examples you saw approximated to this general pattern.



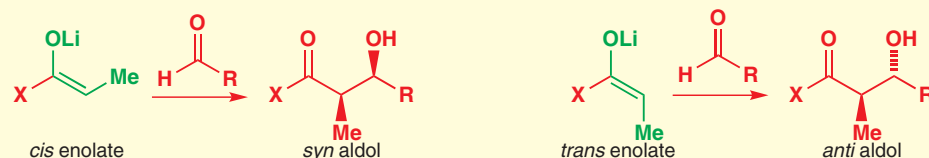
Only one new stereogenic centre is created, so there is no question of diastereoselectivity. But with substituted enolates, two new stereogenic centres are created and we need to be able to predict which diastereoisomer will be formed. Here is an example from p. 626. We did not consider stereochemistry at that stage, but we can now reveal that the *syn* diastereoisomer is the major product of the reaction.



The important point about substituted enolates is that they can exist as two geometrical isomers, *cis* or *trans*. Which enolate is formed is an important factor controlling the diastereoselectivity because it turns out that, in many examples of the aldol reaction, *cis* enolates give *syn* aldols preferentially and *trans* enolates give *anti* aldols preferentially.

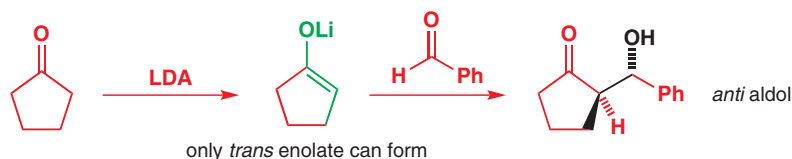
● Diastereoselectivity in aldol reactions

Generally (but certainly not always!) in aldol reactions:



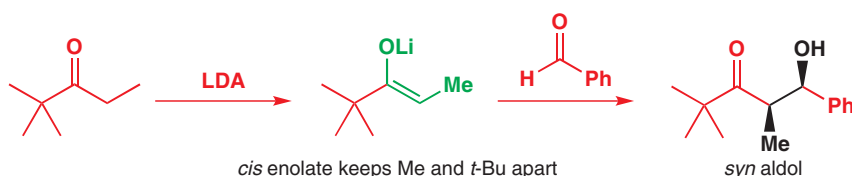
■ This is a very general rule and there are many exceptions—the enolates of some metals [Sn(II), Zr, Ti] give *syn* aldols regardless of enolate geometry. Some related reactions are discussed in Chapter 41.

Let's start by showing some examples and demonstrating how we know this to be the case. Some enolates can exist only as *trans* enolates because they are derived from cyclic ketones. This enolate, for example, reacts with aldehydes to give only the *anti* aldol product.



Interactive mechanism for *anti* stereoselective aldol reaction

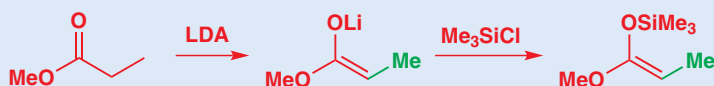
If we choose the group 'X', next to the carbonyl group, to be large, then we can be sure of getting just the *cis* enolate. So, for example, the lithium enolate of this *t*-butyl ketone forms just as one geometrical isomer, and reacts with aldehydes to give only the *syn* aldol product.



Interactive mechanism for *syn* stereoselective aldol reaction

cis and *trans*, *E* and *Z*, *syn* and *anti*

Before going further, there are two points we must clarify. The first is a problem of nomenclature, and concerns the enolates of esters. Here are two closely related ester enolate equivalents, drawn with the same double bond geometry. Is it *E* or *Z*?

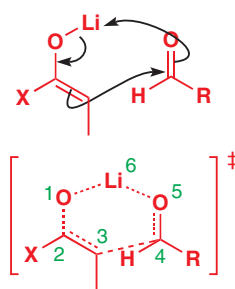


The answer is both! For the Li enolate, the usual rule makes OLi of lower priority than OMe (because Li has a smaller atomic number than C), so it's *E*, while the silyl enol ether (or 'silyl ketene acetal') has OSi of higher priority than OMe (Si has a larger atomic number than C), so it's *Z*. This is merely a nomenclature problem, but it would be irritating to have to reverse all our arguments for lithium and boron enolates (as opposed to, say, tin or silicon ones). So, for the sake of consistency, it is much better to avoid the use of *E* and *Z* with enolates and instead use *cis* and *trans*, which then always refer to the relationship between the substituent and the anionic oxygen (bearing the metal).

The other point concerns *syn* and *anti*. We said earlier that there is no precise definition of these terms: they are a useful way of distinguishing two diastereoisomers provided the structure of at least one of them is presented in diagrammatic form. For aldol products the convention is that *syn* or *anti* refers to the enolate substituent (the green Me in the last example) and the new hydroxyl group, provided the main chain is in the plane of the paper, the way we have encouraged you to draw molecules.

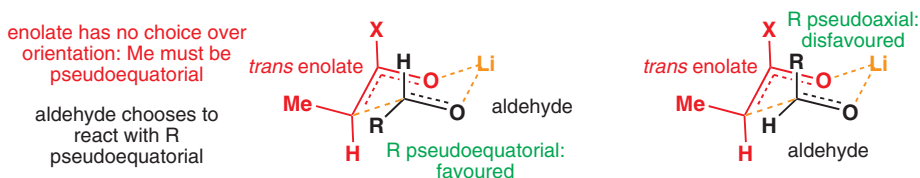
The aldol reaction has a chair-like transition state

These are the experimental facts: how can we explain them? Aldol reactions are another class of stereoselective process with a cyclic transition state. During the reaction, the lithium is transferred from the enolate oxygen to the oxygen of the carbonyl electrophile. This is represented in the margin both in curly arrow terms and as a transition state structure. A six-membered ring is involved, and we can expect this ring to adopt more or less a chair

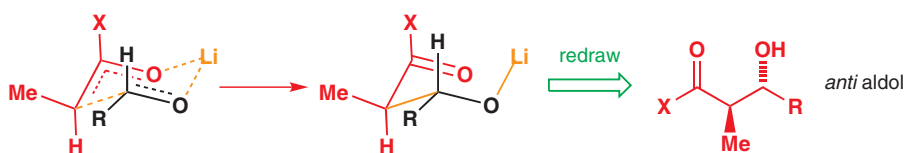


conformation. The easiest way to draw this is first to draw the chair, and then convert atoms to O or Li as necessary. Here it is.

The six-membered ring transition state for the aldol reaction was proposed by Zimmerman and Traxler and is sometimes called the **Zimmerman–Traxler transition state**.



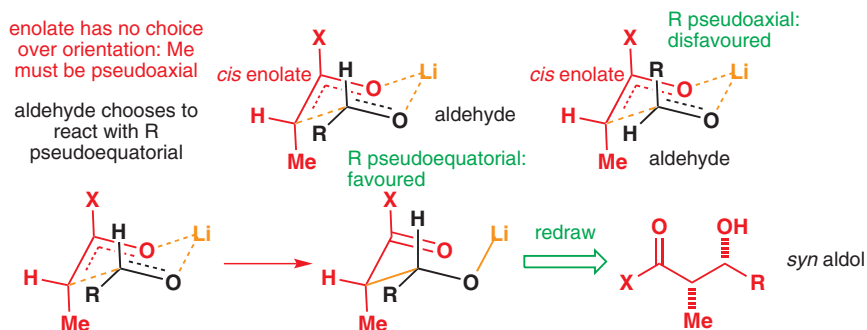
In drawing this chair, we have one choice: do we allow the aldehyde to place R equatorial or axial? Both are possible but, as you should now expect, there are fewer steric interactions if R is equatorial. Note that the enolate doesn't have the luxury of choice. If it is to have three atoms in the six-membered ring, as it must, it can do nothing but place the methyl group pseudoequatorial. The aldol formed from the favoured transition state structure, with R pseudoequatorial, is shown below—first in the conformation of the transition state, and then flattened out on to the page, and it is *anti*.



■ Our advice on p. 860, which we follow again here, was to draw the product first of all in the conformation of the starting material and only then to flatten it out to a 'normal' structure.

Interactive mechanism: *trans* enolate gives *anti* stereoselective aldol reaction

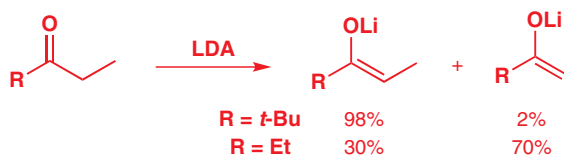
We can do the same for the *cis* enolate. The enolate has no choice but to put its methyl substituent pseudoaxial, but the aldehyde can choose either pseudoequatorial or pseudoaxial. Again, pseudoequatorial is better and the reaction gives the product shown—the *syn* aldol.



Interactive mechanism: *cis* enolate gives *syn* stereoselective aldol reaction

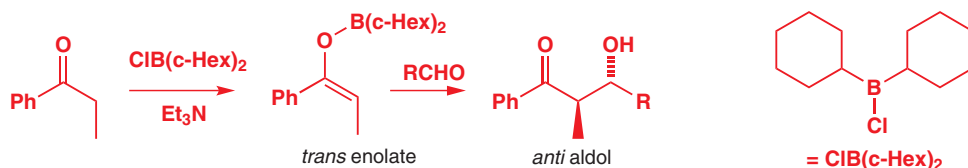
Stereoselective enolization is needed for stereoselective aldols

The cyclic transition state explains how enolate geometry controls the stereochemical outcome of the aldol reaction. But what controls the geometry of the enolate? For lithium enolates of ketones the most important factor is the size of the group that is not enolized. Large groups force the enolate to adopt the *cis* geometry; small groups allow the *trans* enolate to form. Because we can't separate the lithium enolates, we just have to accept that the reactions of ketones with small R will be less diastereoselective.



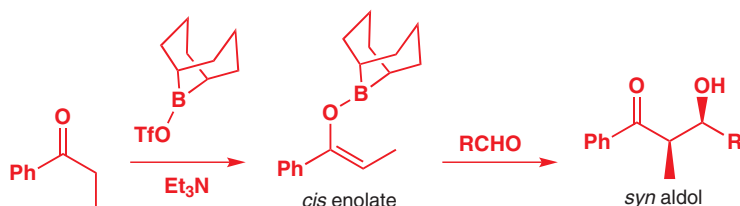
With *boron* enolates, we don't have to rely on the structure of the substrate—we choose the groups on boron—and we can get either *cis* or *trans* depending on which groups these are. Boron enolates are made by treating the ketone with an amine base (often Et₃N or *i*-PrNEt₂) and R₂B–X, where X[–] is a good leaving group such as chloride or triflate (CF₃SO₃[–]). With bulky

groups on boron, such as two cyclohexyl groups, a *trans* enolate forms from most ketones. The boron enolate reacts reliably with aldehydes to give *anti* aldol products through the same six-membered transition state that you saw for lithium enolates.



With smaller B substituents, the *cis* enolate forms selectively. Here, the boron is part of a bicyclic structure known as 9-BBN (9-borabicyclononane). The bicyclic part may look large but, as far as the rest of the molecule is concerned, it's 'tied back' behind the boron, and the methyl group can easily lie *cis* to oxygen. The *cis* enolate then gives *syn* aldol products. Di-*n*-butylboron triflate (Bu_2BOTf) also gives *cis* enolates.

➔ 9-BBN was mentioned in Chapter 19.



● Summary: How to make *syn* and *anti* aldols

To make *syn* aldols of ketones:

- use boron enolate with 9-BBN-OTf or Bu_2BOTf
- from a ketone RCOEt with bulky R, use lithium enolate

To make *anti* aldols of ketones:

- use boron enolate with $(\text{c-Hex})_2\text{BCl}$
- from a cyclic ketone, use lithium enolate

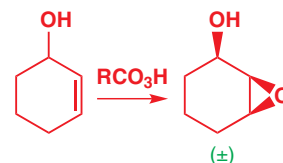
Single enantiomers from diastereoselective reactions

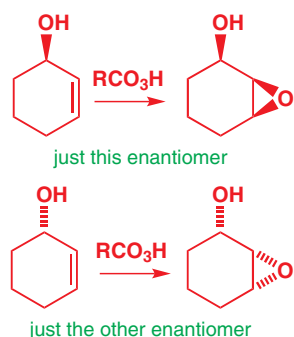
The aldol reactions in the last section made single diastereoisomers from two achiral compounds. No enantiomerically pure reagents were used, so the reaction had no choice but to give the product diastereoisomer as a racemic mixture of its two enantiomers.

In all the other diastereoselective reactions in this chapter, the starting material has been chiral, with the formation of new chiral centres controlled by the configuration of the starting material. Whatever the diastereoselectivity of the reaction, if the starting material is racemic, so will be the product; if the starting material is enantiomerically pure, so will be the product. The epoxidation of the allylic alcohol in the margin illustrates this point.

The reaction starts with racemic material (no stereochemistry is shown at the chiral centre) and makes a racemic product. Of course we have only *drawn* one enantiomer—the only way to draw one diastereoisomer is to choose one enantiomer and draw that—but the indication '±' underneath tells you to expect an equal amount of the other enantiomer as well. Even without this indication, you should be able to work out, in any given case, whether a compound is racemic or not, providing you know where it comes from. Here the starting material is racemic and the reagent is achiral so the product must be racemic.

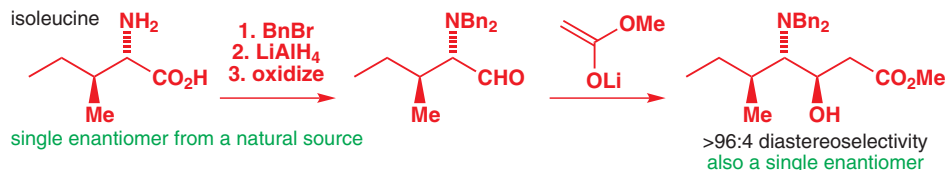
The example of this reaction earlier in the chapter (p. 856) was this type of reaction. The starting alcohol was racemic and the product was just one racemic diastereoisomer—the all *cis* compound. But if the starting material *had* been enantiomerically pure, so would the product. One enantiomer gives one enantiomer of the product: the other enantiomer of the alcohol





gives the other. Both products are the same diastereoisomer (all *cis*) but they are mirror images of each other. If you start with enantiomerically pure compounds, the products will be enantiomerically pure as well.

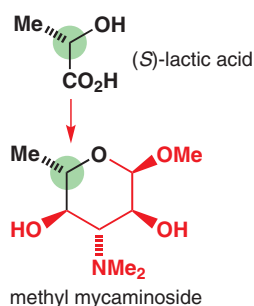
We gave an example of this during the discussion of the Felkin–Anh model. The starting material was the natural amino acid isoleucine and was the enantiomer shown. The product of the aldol reaction was therefore also a single enantiomer. The original chiral centre in both these examples is not affected by the reaction and remains unchanged.



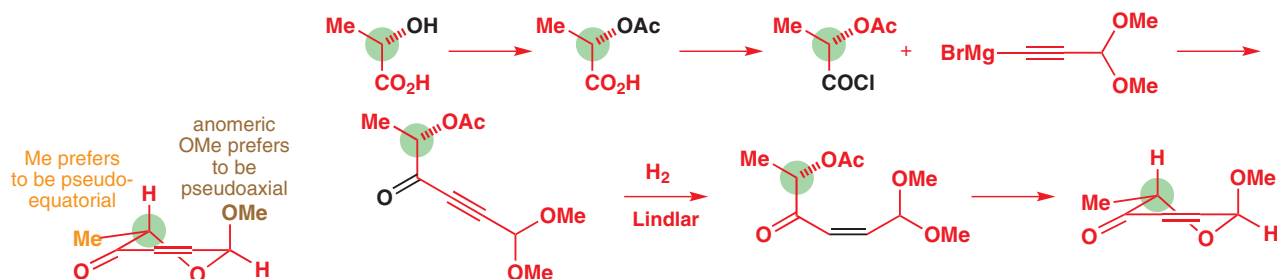
It is much more useful to make enantiomerically pure as well as diastereoisomerically pure compounds, particularly in the synthesis of a drug. The strategy used here is to make the starting material from an enantiomerically pure compound available from nature: in this case an amino acid. These available enantiomerically pure compounds are known collectively as the **chiral pool**. You can read more about this in Chapter 41 on asymmetric synthesis.

If you are making an enantiomerically pure compound with more than one stereogenic centre, only one needs be borrowed from the chiral pool, provided diastereoselective reactions can be used to introduce the others with control over relative stereochemistry. Because the first chiral centre has defined absolute configuration, any diastereoselective reaction that controls the relative stereochemistry of a new chiral centre also defines its absolute configuration.

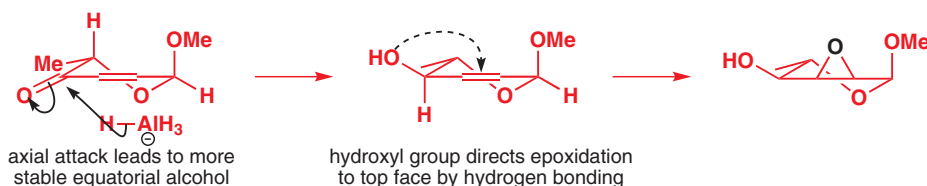
We'll use as an illustration a synthesis of a rare sugar, methyl mycaminoside, containing five chiral centres. Only one chiral centre comes directly from the chiral pool—the rest are introduced diastereoselectively. The naturally derived, enantiomerically pure compound used as the starting material is (*S*)-lactic acid. The starting chiral centre, preserved right through the sequence, is ringed in green.



The ring was built up using familiar chemistry from acetylated (*S*)-lactic acid, and a cyclization step introduced the second chiral centre in the final step of the scheme below. The methyl group goes pseudoequatorial on the newly formed ring, while the anomeric effect, which was explained on p. 801 of Chapter 31, induces the methoxy group to prefer the pseudoaxial position.

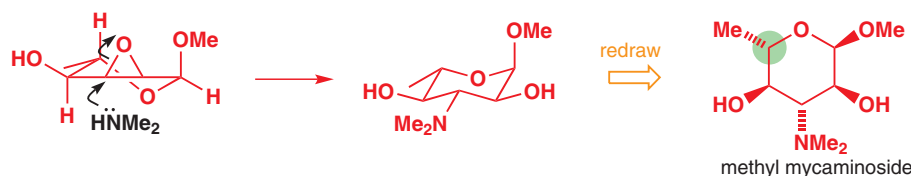


The third stereogenic centre was controlled by reduction of the ketone from the axial direction to give the equatorial alcohol, which then directed introduction of the fourth and fifth stereogenic centres by epoxidation.



➡ The conformational factors governing reduction of cyclohexanones are discussed in Chapter 16. The directing effects of OH groups in epoxidation are discussed in Chapters 32 and earlier in this chapter.

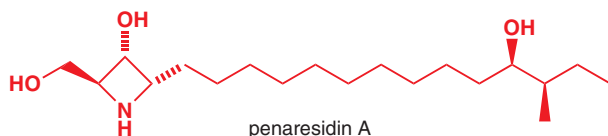
Finally, the simple nucleophilic amine Me_2NH attacks the epoxide with inversion of configuration to give methyl mycaminoside. The conformational drawing shows that all substituents are equatorial except the MeO group, which prefers to be axial because of the anomeric effect. Starting from an enantiomerically pure compound containing one chiral centre, four new chiral centres are introduced in sequence by diastereoselective reactions of various kinds. The final product is necessarily a single enantiomer.



■ Normally, axial attack occurs on cyclohexane epoxides, as explained in Chapter 32, but the rule is not rigid, as you can see here. Equatorial attack occurs because the transition state already has much of the stability of the equatorially substituted product. You should continue to assume that related epoxides will typically undergo axial attack.

The structure and synthesis of penaresidin

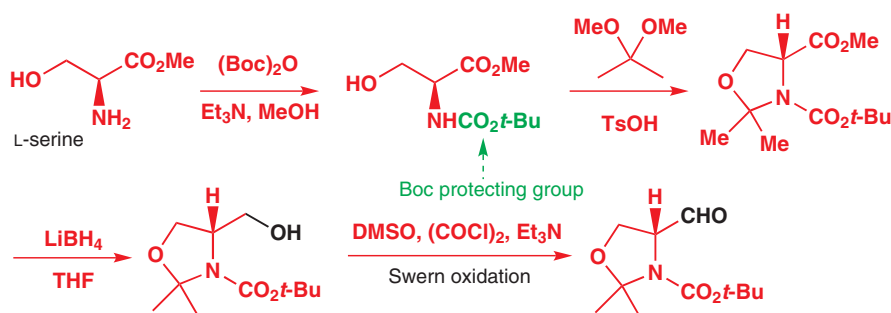
Our last example is a natural product called penaresidin A. It was isolated from a Japanese sponge in 1991, and is now known to have the structure shown below. When it was first discovered, it proved difficult to find out the stereochemistry and, in particular, the relative stereochemistry between the two remotely related groups of chiral centres was not initially known.



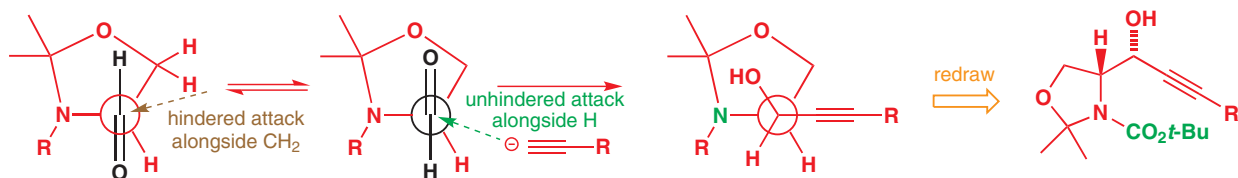
What is sure is the relative stereochemistry around the four-membered azetidine ring: the NMR methods described in Chapter 31 give that. What is also certain is that natural penaresidin A is enantiomerically pure. What Mori and his co-workers set out to do was to make, using unambiguous stereoselective methods, the possible diastereoisomers of penaresidin A to discover which was the same as the natural product.

The challenge of constructing the three chiral centres at the left-hand end of the molecule can be solved by taking just one of them from a natural source—in this case the amino acid L-serine. The amino group of serine was protected as the Boc derivative, and the hydroxyl and amino groups condensed with the dimethoxyacetal of acetone to form a five-membered ring. Now the free ester could be reduced with LiBH_4 and oxidized to the aldehyde by the Swern method (Chapter 27).

➡ The Boc protecting group was introduced on p. 557; the Swern oxidation is on p. 667.

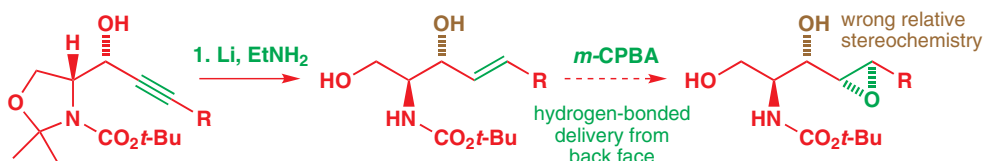


How will this aldehyde react with nucleophiles such as lithiated alkynes? Consider a Felkin–Anh transition state: again, we know that the substituted nitrogen atom, being electronegative and bulky, will lie perpendicular to the carbonyl group in the most reactive conformation. Looking at the two alternatives shown below, it's easy to see that the one on the right allows unhindered attack, and in the synthesis an alkynyl anion was used to make the product shown.



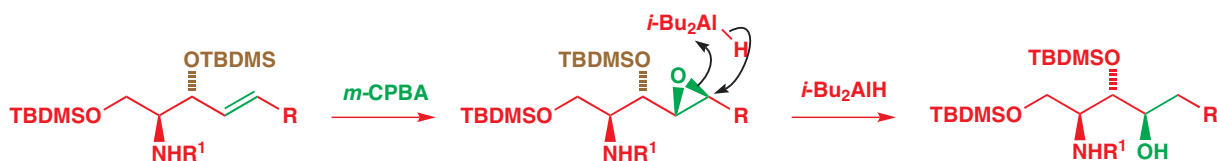
➡ Reduction of alkynes to *E* double bonds is covered in Chapter 27, p. 681.

The alkyne was then reduced to an *E* alkene by a dissolving metal reduction, a step which also hydrolysed the five-membered heterocycle. The next step, an epoxidation, is needed to install the third of the chiral centres at the left-hand end of penarisidine. However, hydrogen-bond directed epoxidation of this allylic alcohol would be expected to give the *syn* product shown, which has the wrong relative stereochemistry between the brown OH group and the epoxide.

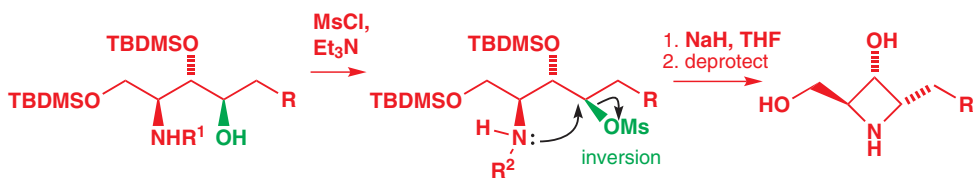


■ The selectivity of the epoxidation of the *E* double bond was still only moderate (about 60:40). From the discussion on p. 866 you should be able to deduce why.

The solution is to use a large blocking group to prevent this brown OH group hydrogen bonding to the *m*-CPBA. The *t*-butyldimethylsilyl group (TBDMS) is the best, and when both OH groups are protected, some of the right diastereoisomer is formed by attack of *m*-CPBA on the top face of the alkene. Reduction of the epoxide with DIBAL (*i*-Bu₂AlH) now gives the correct diastereoisomer.



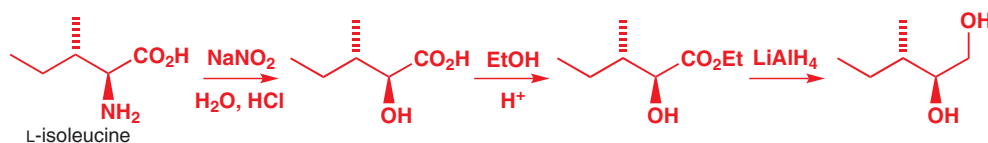
To close the ring, the green hydroxyl group was converted to a good leaving group, mesylate (MeSO₃), ready for an attack by the nitrogen with inversion on treatment with base. Make sure you can see how inversion at this centre gives the stereochemistry shown! The chemists knew at this stage they were on the right track with regard to relative stereochemistry because the NMR spectrum of structures containing any long alkyl chain R were very similar to that of the natural compound.



Confirming the stereochemistry by synthesis

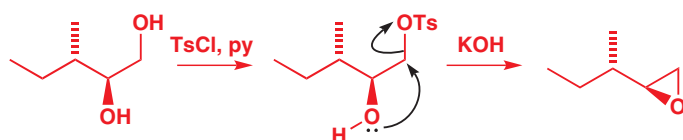
The other two chiral centres at the right-hand end of the chain are so far removed from the ring (by 10 CH₂ groups) that there is no simple way to determine their stereochemistry relative to the three at the left-hand end by NMR. The solution to problems of assignment like this is often to make the various isomers by unambiguous synthesis and compare the NMR spectra of the natural and synthetic compounds. This is what Mori did in this case.

The chiral pool can again be called into play by using another amino acid, L-isoleucine, as starting material. First the amino group must be converted to a leaving group by diazotization with nitrous acid (sodium nitrite in dilute HCl) and substituted by water to give a hydroxy acid. The acid is esterified and reduced to a diol.

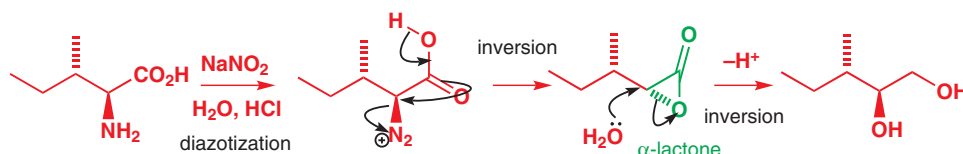


➡ We used NaNO_2 in acid to convert amino groups into diazonium salts containing the excellent leaving group N_2 in Chapter 22, p. 520.

Conversion of the diol's primary hydroxyl group to a leaving group (here a tosylate) allows the epoxide to be formed with retention of the two stereogenic centres of the starting material. Cyclization in base gives an epoxide. Overall, the enantiomerically pure starting material is converted stereospecifically into a single enantiomer of a single diastereoisomer of the epoxide.

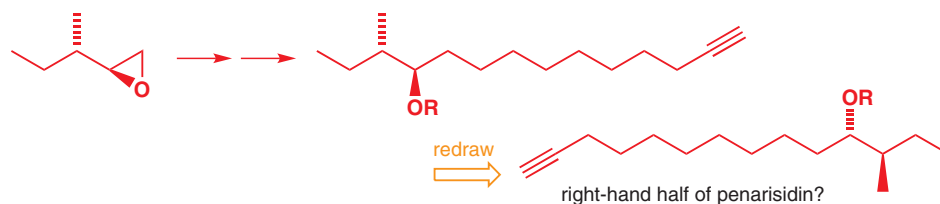


Before we go on, look back at the first reaction of this sequence—the conversion of L-leucine to the hydroxy acid. The stereochemistry may surprise you: look carefully and you will see that the amino group has been displaced with *retention* of stereochemistry. Retentive substitutions usually indicate double inversions, and here the carboxylic acid gets involved in the displacement to give (with inversion) a very unstable compound called an α -lactone whose strained ring is opened by water, also with inversion.

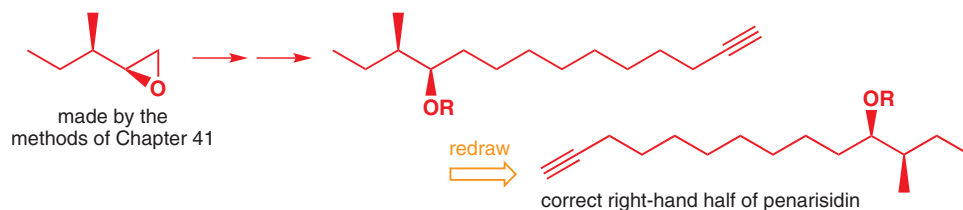


➡ We will address similar examples of 'neighbouring group participation' in Chapter 36. There is more on α -lactones on p. 934. The mechanism for the diazotization step is in Chapter 22, p. 521.

The epoxide may now be opened with a nucleophile to give the right-hand half of the target molecule. The alkyne shown below, which has an *anti* relationship between the hydroxyl and methyl groups, was made and linked to the left-hand half of penaresidin A by using it to attack the aldehyde the method described above. However, the final product was not the same as natural penaresidin A!



Clearly, some aspect of relative stereochemistry was wrong. So the synthesis was repeated, this time using the *syn* diastereoisomer of the substituted alkyne obtained using one of the methods we will describe in Chapter 41. With this isomer the final compound had spectroscopic data identical with the natural product, and the question of its stereochemistry was solved. It is not uncommon for synthesis to be the only reliable way of proving the detailed structure of a compound.



Looking forward

Once you have got hold of a molecule as a single enantiomer, however simple that molecule may be, you can always use reliably diastereoselective reactions of the type described in this chapter and the one before to decorate it with further chiral centres. This is a very important point that underlies the field of asymmetric synthesis, which we will cover in Chapter 41. There you will see developments of the ideas we have just been describing, where chiral centres derived from nature are used to introduce new stereochemistry even though they themselves need not appear in the final product. But before we move on to such reactions, we need to cover a handful of important new reaction mechanisms, many of which provide further ways of introducing new stereochemical features into molecules. The first of these new classes of reactions is *cycloadditions*.

Further reading

For explanations of pericyclic reactions and other reactions, using the full molecular orbital treatment, consult: Ian Fleming, *Molecular Orbitals and Organic Chemical Reactions, Student Edition*, Wiley, Chichester 2009. There is also a more comprehensive edition intended for practicing chemists, called the *Library Edition*.

For a comprehensive treatment of diastereoselectivity and the chiral pool approach to asymmetric synthesis as well as control of double bond geometry see P. Wyatt and S. Warren, *Organic*

Synthesis: Strategy and Control, Wiley, Chichester, 2007 and the accompanying *Workbook*, also Wiley, 2008.

Leading references for the synthesis of penaresidin are K. Mori and group, *J. Chem. Soc., Perkin Trans. 1*, 1997, 97; S. Knapp and Y. Dong, *Tetrahedron Lett.*, 1997, **38**, 3813 and H. Yoda and group, *Tetrahedron Lett.*, 2003, **44**, 977. The synthesis of methyl mycaminoside is from Koga, K., Yamada, S.-I., Yoh, M., Mizoguchi, T. *Carbohydr. Res.* **1974**, **36**, C9–C11.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>